

Project Abstract

This project presents a 2 degrees of freedom (DoF) self-adaptive bionic ankle designed for humanoid robots to navigate uneven terrain. Some potential applications for humanoid robots includes exploration projects in rough terrain such as the surface of Mars and rescue projects in complex environment such as a collapsed building.

In addition to predecessors' spring mechanism, this new bionic ankle has an additional motor is needed to control the leg's dorsiflexion and plantar flexion and an ankle's inversion and eversion movement, a self-adaptive program uses a combination of IMU and laser distance sensors to aid walking on uneven ground, and strings to put the motors near the leg instead of around the ankle.

For its structure, it uses two motors, simulating as muscles to drive the leg, a universal joint to act as the natural joint of a human ankle and contains a spring system that can absorb shock, paralleling to the articular cartilage and ligaments. It is capable of dorsiflexion up to 24.1° , plantar flexion up to 33.7° , inversion up to 21.9° , and eversion up to 14.6° . Using additional distance sensors, it can calculate the pitch and roll of the foot relative to the ground, and it can move the foot to become parallel to the ground using the PID controller algorithm. This allows it to balance itself on uneven terrain and swiftly land perfectly on inclined areas with a delay of 25 millisecond.